Class 14: RNA-Seq analysis mini-project

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Background

Data Import

We have 2 ket input files: counts and metadata

```
countData <- read.csv("GSE37704_featurecounts.csv", row.names=1)
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
```

```
ENSG00000273547     939         0         0         0         0         0
ENSG00000187634    3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

```
colData <- read.csv("GSE37704_metadata.csv", row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

We need to remove the first length" column from `countData` to have a 1:! corespondance with `colData` rows.

```
rownames(colData) == colnames(countData)
```

```
Warning in rownames(colData) == colnames(countData): longer object length is
not a multiple of shorter object length
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

Check and Tidy

> Q. Complete the code below to remove the troublesome first column from countData

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

> Q2. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```r
countData = countData[rowSums(countData) > 0,]
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

Remove zero counts genes

Some genes (rows) have no count data (i.e. zero values). We should remove these before any further analysis.

```r
to.keep <- rowSums(countData) > 0
countData <- countData[to.keep, ]
```

Run DESeq

```r
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics


Attaching package: 'generics'


The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union



Attaching package: 'BiocGenerics'


The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs


The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min



Attaching package: 'S4Vectors'


The following object is masked from 'package:utils':

    findMatches


The following objects are masked from 'package:base':

    expand.grid, I, unname


Loading required package: IRanges
```

```
Attaching package: 'IRanges'

The following object is masked from 'package:grDevices':

    windows

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase
```

```
Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                       colData = colData,
                       design = ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```
dds= DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

Results

```r
res <- results(dds)
res
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 15975 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
...                   ...            ...       ...        ...         ...
ENSG00000273748  35.30265       0.674387  0.303666   2.220817 2.63633e-02
ENSG00000278817   2.42302      -0.388988  1.130394  -0.344117 7.30758e-01
ENSG00000278384   1.10180       0.332991  1.660261   0.200565 8.41039e-01
ENSG00000276345  73.64496      -0.356181  0.207716  -1.714752 8.63908e-02
ENSG00000271254 181.59590      -0.609667  0.141320  -4.314071 1.60276e-05
                       padj
                  <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76549e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
...                     ...
ENSG00000273748 4.79091e-02
ENSG00000278817 8.09772e-01
ENSG00000278384 8.92654e-01
ENSG00000276345 1.39762e-01
ENSG00000271254 4.53648e-05
```

Get Results

> Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

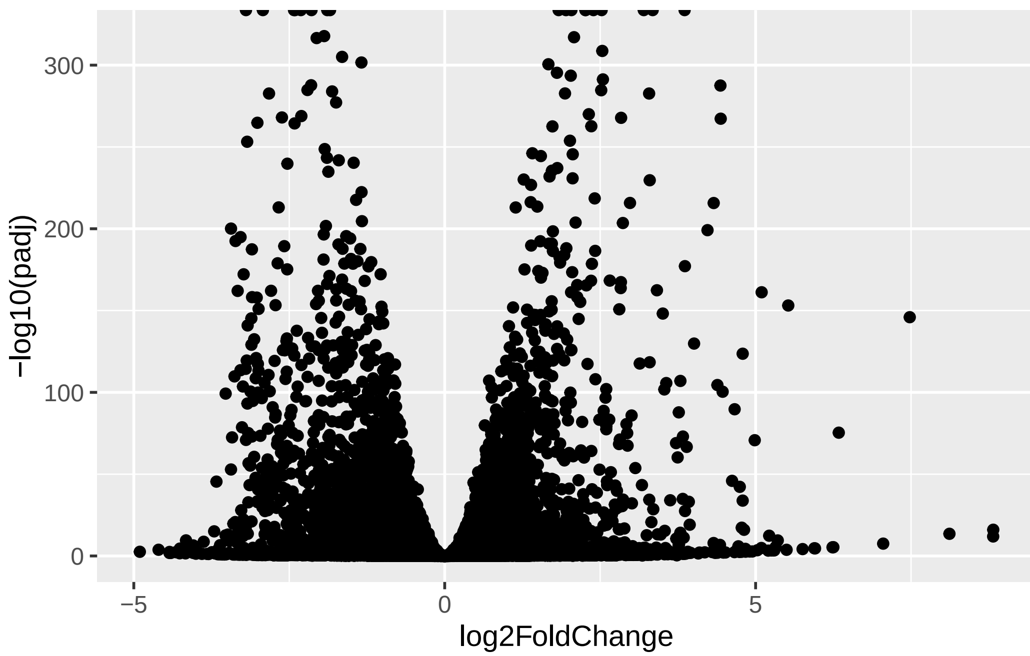
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano plot

```r
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log10(padj)) +
  geom_point()
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



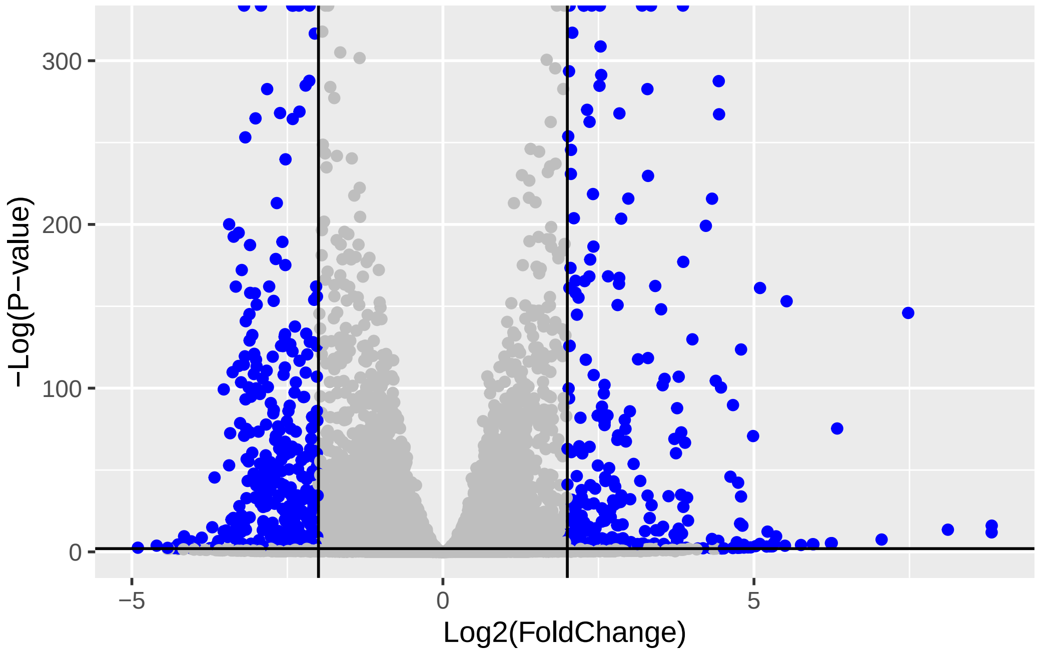
> Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

Let's add some color to this plot along with cutoff lines for fold-change and P-value

```
mycols <-  rep ("gray", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "blue"
mycols[res$padj > 0.01] <- "gray"

ggplot(res) +
  aes(x = log2FoldChange,
      y = -log10(padj)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log10(0.01))
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add Annotation

> Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

MapIds

```r
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```r
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```r
res$symbol <- mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <-  mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$genename <- mapIds(org.Hs.eg.db,
                       keys = row.names(res),
                       keytype = "ENSEMBL",
                       column = "GENENAME",
                       multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res <- as.data.frame(res)
```

> Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project director

```r
res = res[order(res$padj),]
write.csv(res, file="deseq_results.csv")
```

Pathway Analysis

```r
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
library(gage)
```

```r
library(gageData)
```

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
# Remove NA Entrez IDs
foldchanges <- foldchanges[!is.na(names(foldchanges))]
```

```r
data(kegg.sets.hs)
data(sigmet.idx.hs)
```

```r
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```r
# Look at the first few down (less) pathways
head(keggres$less)
```

```
                                              p.geomean stat.mean
hsa04110 Cell cycle                        9.366328e-06 -4.369134
hsa03030 DNA replication                   9.592718e-05 -3.946505
hsa05130 Pathogenic Escherichia coli infection 1.440011e-04 -3.758434
hsa03013 RNA transport                     1.439613e-03 -3.014203
hsa03440 Homologous recombination          3.106306e-03 -2.848189
hsa04114 Oocyte meiosis                    3.888649e-03 -2.688749
                                                  p.val       q.val
hsa04110 Cell cycle                        9.366328e-06 0.001966929
```

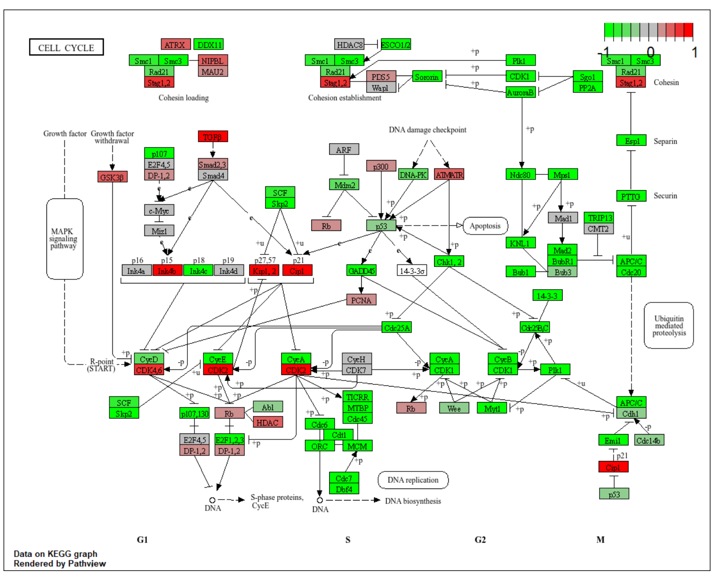
```
hsa03030 DNA replication                          9.592718e-05 0.010072354
hsa05130 Pathogenic Escherichia coli infection 1.440011e-04 0.010080080
hsa03013 RNA transport                            1.439613e-03 0.075579703
hsa03440 Homologous recombination                 3.106306e-03 0.130464871
hsa04114 Oocyte meiosis                           3.888649e-03 0.136102700
                                               set.size         exp1
hsa04110 Cell cycle                                 121 9.366328e-06
hsa03030 DNA replication                             36 9.592718e-05
hsa05130 Pathogenic Escherichia coli infection       53 1.440011e-04
hsa03013 RNA transport                              144 1.439613e-03
hsa03440 Homologous recombination                    28 3.106306e-03
hsa04114 Oocyte meiosis                             102 3.888649e-03
```

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory C:/Users/user/Documents/BIMM 143/Class14
```

```
Info: Writing image file hsa04110.pathview.png
```



Go Analysis

Focus on the Biological Process "BP"

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
data(go.sets.hs)
data(go.subs.hs)

gobpsets = go.sets.hs[go.subs.hs$BP]
```

```
gobpres = gage(foldchanges, gsets=gobpsets)
```

```
head(gobpres$greater)
```

```
                                           p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.132355e-05  3.836423 8.132355e-05
GO:0002009 morphogenesis of an epithelium 1.301491e-04  3.672346 1.301491e-04
GO:0048729 tissue morphogenesis           1.324453e-04  3.663702 1.324453e-04
GO:0007610 behavior                       1.786592e-04  3.585306 1.786592e-04
GO:0035295 tube development               5.558494e-04  3.273391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  5.613074e-04  3.277405 5.613074e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1804788      113 8.132355e-05
GO:0002009 morphogenesis of an epithelium 0.1804788      339 1.301491e-04
GO:0048729 tissue morphogenesis           0.1804788      424 1.324453e-04
GO:0007610 behavior                       0.1825897      426 1.786592e-04
GO:0035295 tube development               0.3325708      391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  0.3325708      257 5.613074e-04
```

```
head(gobpres$less)
```

```
                                          p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.735928e-15 -8.047667 1.735928e-15
GO:0000280 nuclear division              4.808128e-15 -7.923688 4.808128e-15
GO:0007067 mitosis                       4.808128e-15 -7.923688 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.312717e-14 -7.781713 1.312717e-14
GO:0007059 chromosome segregation        2.149699e-11 -6.868718 2.149699e-11
GO:0000236 mitotic prometaphase          1.794819e-10 -6.689146 1.794819e-10
                                                q.val set.size         exp1
```

```
GO:0048285 organelle fission             6.551876e-12      376 1.735928e-15
GO:0000280 nuclear division              6.551876e-12      352 4.808128e-15
GO:0007067 mitosis                       6.551876e-12      352 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.341596e-11      362 1.312717e-14
GO:0007059 chromosome segregation        1.757594e-08      142 2.149699e-11
GO:0000236 mitotic prometaphase          1.222870e-07       84 1.794819e-10
```

OPTIONAL Identify top significant genes

```r
res_sig <- res[which(res$padj < 0.01 & abs(res$log2FoldChange) > 2), ]
head(res_sig, 10)
```

```
                 baseMean log2FoldChange      lfcSE      stat pvalue padj
ENSG00000117519  4483.627      -2.422719 0.06000162 -40.37756      0    0
ENSG00000183508  2053.881       3.201955 0.07241720  44.21540      0    0
ENSG00000159176  5692.463      -2.313738 0.05755337 -40.20160      0    0
ENSG00000164251  2348.770       3.344508 0.06907176  48.42078      0    0
ENSG00000124766  2576.653       2.392288 0.06170862  38.76749      0    0
ENSG00000188153  2944.126       2.266080 0.05526813  41.00158      0    0
ENSG00000122861 28007.143       2.262525 0.05521833  40.97417      0    0
ENSG00000148773  1840.197      -3.196058 0.07578504 -42.17268      0    0
ENSG00000139289 12942.911       2.037362 0.04629734  44.00602      0    0
ENSG00000211448  1884.757       2.523722 0.06539813  38.59012      0    0
                symbol entrez                                        genename
ENSG00000117519   CNN3   1266                                      calponin 3
ENSG00000183508 TENT5C  54855                terminal nucleotidyltransferase 5C
ENSG00000159176  CSRP1   1465                cysteine and glycine rich protein 1
ENSG00000164251  F2RL1   2150                        F2R like trypsin receptor 1
ENSG00000124766   SOX4   6659                     SRY-box transcription factor 4
ENSG00000188153 COL4A5   1287                     collagen type IV alpha 5 chain
ENSG00000122861   PLAU   5328                  plasminogen activator, urokinase
ENSG00000148773  MKI67   4288                      marker of proliferation Ki-67
ENSG00000139289 PHLDA1  22822 pleckstrin homology like domain family A member 1
ENSG00000211448   DIO2   1734                          iodothyronine deiodinase 2
```